Karim khalil

(AI Engineer)

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Professional Summery

AI Engineer with experience in developing AI-powered applications, machine learning solutions, and data-driven systems. Skilled in Python, SQL, LLM applications, and workflow automation. Experienced in platform operations, data analysis, dashboard validation, and business process improvement, with a passion for building practical AI solutions that improve operational efficiency.

Experience

Platfarm | Royal Commission for AlUla (RCU) — AlUla, Saudi Arabia | On-site *June 2026 - Present*
Platform Operation & Data Analyst

- Managed platform operations while ensuring data accuracy and system reliability.
- Validated dashboards and reports to ensure data quality and consistency.
- Collaborated with stakeholders to gather requirements and improve business processes.
- Built AI applications and automated workflows using Python, Replit, OpenClaw, and LLM technologies.

Nasam Technoloy – Raidah, Saudi Arabia | Remote *May 2025 - April 2026*
AI Engineer

- Built and deployed AI-powered APIs using FastAPI, enabling internal systems to consume real-time inference results efficiently.
- Optimized real-time video processing pipelines for environmental monitoring use cases, improving response latency by ~15%.
- Tested and validated AI services using Postman, ensuring consistent performance and reliability across deployments.

Omni Clouds – Dubai, United Arab Emirates | Remote *July 2024 - February 2025*
AI Engineer

- Developed and deployed computer vision models for real-time video monitoring systems, containerized using Docker for internal production use.

- Implemented YOLO-based object detection pipelines for live video streams, improving detection reliability and reducing false alerts by ~20%.
- Collaborated with software engineers to integrate AI models into internal systems and applications, supporting continuous real-time inference and system stability.

Technical Skills

- Programming languages: Python, SQL
- AI & Machine Learning: Machine Learning, Deep Learning, Large Language Models (LLMs), AI Applications, Computer Vision
- Data Analytics: Data Analysis, Data Cleaning, Data Validation, Business Intelligence, Dashboard Validation, Data Visualization, Excel
- Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas
- Tools & Technologies: OpenClaw, Replit, FastAPI, REST APIs, Docker, Git, GitHub, Postman

Education

CIC - Canadian International College
Bachelor of Science in Computer Science *2024*

Certification

Data analysis - egFWD

AI for everyone - Coursera

Competitve programming level 1 - Coach Academy EG

Geographic Information Services Professional (GIS)-NARSS

communication skills - CIC - Canadian International College

Project

AI-Powered WhatsApp Reporting Assistant

- Developed an AI-powered WhatsApp application using **OpenClaw** and **Replit** to automate report generation and delivery.
- Integrated LLM capabilities to process requests and deliver reports through WhatsApp, improving reporting efficiency.

Driver Inattention Monitoring System

- Developed a real-time driver monitoring system using CNNs and facial landmark tracking to detect fatigue and distraction.
- Optimized inference performance for continuous real-time processing, achieving approximately 20% faster detection responsiveness
- Tools & Tech: Python, PyTorch, OpenCV, Dlib, Multi-threading, NumPy

Human Activity Detection System

- Built activity recognition models using transfer learning on temporal video data for elderly

safety and posture monitoring.

- Delivered real-time inference suitable for continuous surveillance and alerting.
- Tools & Tech: Python, TensorFlow, Keras, OpenCV, Scikit-learn

PPE Detection System

- Implemented YOLOv8-based object detection models to monitor PPE compliance in industrial environments.
- Trained and validated models on custom datasets, reducing manual monitoring by ~30%.
- Tools & Tech: Python, PyTorch, YOLOv8, OpenCV, LabelImg

Restricted Area Intrusion Detection System

- Developed an AI security system using object detection and motion analysis for factory surveillance.
- Generated real-time alerts with visual overlays, improving response time by ~25%.
- Tools & Tech: Python, OpenCV, YOLOv8, NumPy, Flask

Fire and Smoke Detection System

- Engineered a CNN-based classifier for early fire and smoke detection across varied environments.
- Optimized for low-latency deployment in smart surveillance pipelines.
- Tools & Tech: Python, TensorFlow, Keras, OpenCV, Pandas, NumPy